

Операционные системы

Основы интерфейса взаимодействия пользователя с системой Unix на уровне командной строки

Харламова Арина Александровна

2026-09-04

Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Приобретение практических навыков взаимодействия пользователя с системой посредством командной строки.

- 1 Определить имя и путь домашнего каталога
- 2 Изучить команду `ls`.
- 3 Выполнить действия с каталогами.
- 4 Получить дополнительные сведения при помощи справки по командам.
- 5 Изучить команду `history`.

2. Процесс выполнения лабораторной работы

```
aaharlamova@aaharlamova:~$ cd  
aaharlamova@aaharlamova:~$ pwd  
/home/aaharlamova  
aaharlamova@aaharlamova:~$
```

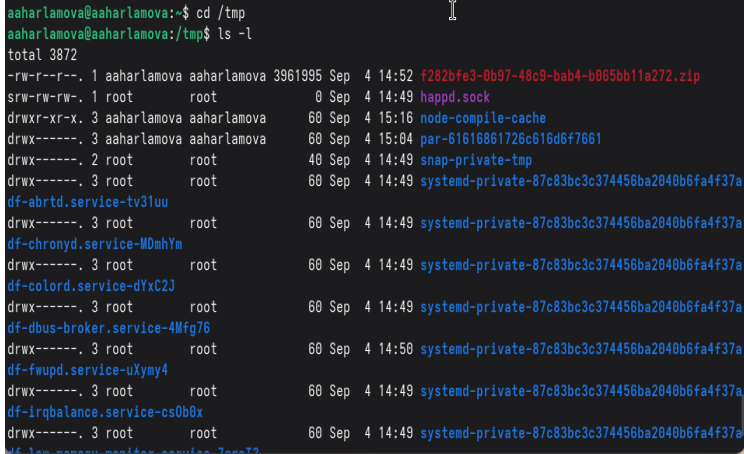
Рисунок 1: Путь к домашнему каталогу

```
aaharlamova@aaharlamova:~$ cd
aaharlamova@aaharlamova:~$ pwd
/home/aaharlamova
aaharlamova@aaharlamova:~$
aaharlamova@aaharlamova:~$ cd /tmp
aaharlamova@aaharlamova:/tmp$ ls
f282bfe3-0b97-48c9-bab4-b065bb11a272.zip
happd.sock
node-compile-cache
par-61616861726c616d6f7661
snap-private-tmp
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-abrt.service-tv31uu
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-chrond.service-MDmhYm
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-colord.service-dYxC2J
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-dbus-broker.service-4Mfg76
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-fwupd.service-uXymy4
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-irqbalance.service-cs0b0x
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-low-memory-monitor.service-7qgsT3
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-ModemManager.service-nq6RXG
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-pcsd.service-uen3a0
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-polkit.service-fVDtuP
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-rtkit-daemon.service-8C0uej
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-switcheroo-control.service-yDQskn
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-systemd-logind.service-R5b9Bx
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-systemd-resolved.service-edhZXt
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-upower.service-jAi5HY
VMwareDnD
vmware-root_895-3979642976
aaharlamova@aaharlamova:/tmp$
```

Рисунок 2: Команда ls


```
aaharlamova@aaharlamova:/tmp$ ls -la
..
f282bfe3-0b97-48c9-bab4-b065bb11a272.zip
.font-unix
happd.sock
.ICE-unix
node-compile-cache
par-61616861726c616d6f7661
snap-private-tmp
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-abrtd.service-tv31uu
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-chronyd.service-MDmhYm
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-colord.service-dYxC2J
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-dbus-broker.service-4Mfg76
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-fwupd.service-uXymy4
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-irqbalance.service-cs0b0x
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-low-memory-monitor.service-7qgsT3
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-ModemManager.service-nqGRXG
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-pcsd.service-uen3a0
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-polkit.service-fVDtuP
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-rtkit-daemon.service-8C0uej
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-switcheroo-control.service-yDQskn
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-systemd-logind.service-R5b9Bx
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-systemd-resolved.service-edhZXt
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-upower.service-jA15HY
VMwareDnD
vmware-root_895-3979642976
.X0-lock
.X11-unix
.X1-lock
.XIM-unix
aaharlamova@aaharlamova:/tmp$
```

Рисунок 3: Команда ls -la



```
aaharlamova@aaharlamova:~$ cd /tmp
aaharlamova@aaharlamova:/tmp$ ls -l
total 3872
-rw-r--r--. 1 aaharlamova aaharlamova 3961995 Sep  4 14:52 f282bfe3-0b97-48c9-bab4-b065bb11a272.zip
srw-rw-rw-. 1 root      root          0 Sep  4 14:49 happd.sock
drwxr-xr-x. 3 aaharlamova aaharlamova   60 Sep  4 15:16 node-compile-cache
drwx-----. 3 aaharlamova aaharlamova   60 Sep  4 15:04 par-61616861726c616d6f7661
drwx-----. 2 root      root          40 Sep  4 14:49 snap-private-tmp
drwx-----. 3 root      root          60 Sep  4 14:49 systemd-private-87c83bc3c374456ba2040b6fa4f37a
df-abrttd.service-tv31uu
drwx-----. 3 root      root          60 Sep  4 14:49 systemd-private-87c83bc3c374456ba2040b6fa4f37a
df-chrond.service-MDmhYm
drwx-----. 3 root      root          60 Sep  4 14:49 systemd-private-87c83bc3c374456ba2040b6fa4f37a
df-colord.service-dYxC2J
drwx-----. 3 root      root          60 Sep  4 14:49 systemd-private-87c83bc3c374456ba2040b6fa4f37a
df-dbus-broker.service-4Mfg76
drwx-----. 3 root      root          60 Sep  4 14:50 systemd-private-87c83bc3c374456ba2040b6fa4f37a
df-fwupd.service-uXymy4
drwx-----. 3 root      root          60 Sep  4 14:49 systemd-private-87c83bc3c374456ba2040b6fa4f37a
df-irqbalance.service-cs0b0x
drwx-----. 3 root      root          60 Sep  4 14:49 systemd-private-87c83bc3c374456ba2040b6fa4f37a
df-udev-monitor.service-7xxT2
```

Рисунок 4: Команда ls -l

Опции команды ls

```
aaharlamova@aaharlamova:/tmp$ ls -f
..
node-compile-cache
par-61616861726c616d6f7661
VMwareDnD
f282bfe3-0b97-48c9-bab4-b065bb11a272.zip
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-fwupd.service-uXymy4
.X1-lock
.X0-lock
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-pcsd.service-uen3a0
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-colord.service-dYxC2J
happd.sock
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-ModemManager.service-nq6RXG
vmware-root_895-3979642976
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-systemd-logind.service-R5b9Bx
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-upower.service-jA15HY
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-switcheroo-control.service-yDQskn
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-rtkit-daemon.service-8C0uej
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-polkit.service-fVDtuP
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-low-memory-monitor.service-7qgsT3
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-chrond.service-MDmhYm
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-irqbalance.service-cs0b0x
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-abrt.service-tv31uu
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-dbus-broker.service-4Mfg76
systemd-private-87c83bc3c374456ba2040b6fa4f37adf-systemd-resolved.service-edhZXt
.font-unix
.XIM-unix
.ICE-unix
.X11-unix
snap-private-tmp
aaharlamova@aaharlamova:/tmp$
```

Рисунок 5: Команда ls -f

```
aaharlamova@aaharlamova:/tmp$  
aaharlamova@aaharlamova:/tmp$ cd /var/spool/  
aaharlamova@aaharlamova:/var/spool$ ls -l  
total 0  
drwxr-x--x. 1 root abrt 24476 Aug 31 13:37 abrt  
drwx-----. 1 abrt abrt    0 Dec  4 2025 abrt-upload  
drwxr-xr-x. 1 root root   66 Oct 23 2025 anacron  
drwx-----. 1 root root   18 Oct 23 2025 at  
drwx-----. 1 root root    0 Aug  6 2025 cron  
drwx--x---. 1 root lp      6 Dec 12 2025 cups  
drwxr-xr-x. 1 root root    0 Jul 30 2025 lpd  
drwxrwxr-x. 1 root mail  298 Sep  4 14:48 mail  
drwxr-xr-x. 1 root root    0 Jan 27 2026 plymouth  
aaharlamova@aaharlamova:/var/spool$
```

Рисунок 6: Каталог /var/spool

```
aaharlamova@aaharlamova:/var/spool$ cd
aaharlamova@aaharlamova:~$ ls
Desktop  Downloads  git-extended  Music  Pictures  Public  Templates  Videos  work
aaharlamova@aaharlamova:~$ ls -al
total 24
drwx----- 1 aaharlamova aaharlamova 370 Sep  4 15:29 .
drwxr-xr-x 1 root        root        298 Sep  4 14:48 ..
-rw----- 1 aaharlamova aaharlamova 3001 Sep  4 15:33 .bash_history
-rw-r--r-- 1 aaharlamova aaharlamova  18 Jul 23 2025 .bash_logout
-rw-r--r-- 1 aaharlamova aaharlamova 144 Jul 23 2025 .bash_profile
-rw-r--r-- 1 aaharlamova aaharlamova 681 Sep  4 15:16 .bashrc
drwx----- 1 aaharlamova aaharlamova 484 Sep  4 15:22 .cache
drwx----- 1 aaharlamova aaharlamova 338 Sep  4 15:21 .config
drwxr-xr-x 1 aaharlamova aaharlamova   0 Sep  4 14:50 Desktop
drwxr-xr-x 1 aaharlamova aaharlamova   0 Sep  4 14:50 Documents
drwxr-xr-x 1 aaharlamova aaharlamova   0 Sep  4 14:50 Downloads
-rw-r--r-- 1 aaharlamova aaharlamova 334 Nov 14 2025 .emacs
-rw-r--r-- 1 aaharlamova aaharlamova 239 Sep  4 15:29 .gitconfig
drwxr-xr-x 1 aaharlamova aaharlamova  74 Sep  4 15:27 git-extended
drwx----- 1 aaharlamova aaharlamova 136 Sep  4 14:57 .gnupg
drwx----- 1 aaharlamova aaharlamova  20 Sep  4 14:50 .local
drwxr-xr-x 1 aaharlamova aaharlamova  34 Oct 23 2025 .mozilla
drwxr-xr-x 1 aaharlamova aaharlamova   0 Sep  4 14:50 Music
drwxr-xr-x 1 aaharlamova aaharlamova   0 Sep  4 14:50 Pictures
drwxr-xr-x 1 aaharlamova aaharlamova   0 Sep  4 14:50 Public
drwx----- 1 aaharlamova aaharlamova 132 Sep  4 15:02 .ssh
drwxr-xr-x 1 aaharlamova aaharlamova   0 Sep  4 14:50 Templates
drwxr-xr-x 1 aaharlamova aaharlamova  18 Sep  4 15:03 .texlive2025
drwxr-xr-x 1 aaharlamova aaharlamova   0 Sep  4 14:50 Videos
drwxr-xr-x 1 aaharlamova aaharlamova  10 Sep  4 15:01 work
aaharlamova@aaharlamova:~$
```

Рисунок 7: Файлы в домашнем каталоге

```
aaharlamova@aaharlamova:~$  
aaharlamova@aaharlamova:~$ mkdir newdir  
aaharlamova@aaharlamova:~$ mkdir newdir/morefun  
aaharlamova@aaharlamova:~$ mkdir letters memos misk  
aaharlamova@aaharlamova:~$ ls  
Desktop  Downloads  letters  misk  newdir  Public  Videos  
Documents  git-extended  memos  Music  Pictures  Templates  work  
aaharlamova@aaharlamova:~$ rm letters/ memos/ misk/  
rm: cannot remove 'letters/': Is a directory  
rm: cannot remove 'memos/': Is a directory  
rm: cannot remove 'misk/': Is a directory  
aaharlamova@aaharlamova:~$ rm -r letters/ memos/ misk/  
aaharlamova@aaharlamova:~$ rm -r newdir/  
aaharlamova@aaharlamova:~$ ls  
Desktop  Documents  Downloads  git-extended  Music  Pictures  Public  Templates  Videos  work  
aaharlamova@aaharlamova:~$
```

Рисунок 8: Действия с каталогами

```
./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_extensions/y
amadharmaminted-quarto':
_extension.yml minted-quarto.lua

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/image':
solvay.jpg

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources':
csl tex

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources/cs
l':
gost-r-7-0-5-2008-numeric.csl

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources/te
x':
preamble.tex

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/scripts':
image-report mpv-shot
aaharlamova@aaharlamova:~$ ls -t
git-extended work Desktop Documents Downloads Music Pictures Public Templates Videos
aaharlamova@aaharlamova:~$
```

Рисунок 9: Команда ls -R и ls -t

```
aaharlamova@aaharlamova:~$ help cd
cd: cd [-L|[-P [-e]] [-@]] [dir]
    Change the shell working directory.

Change the current directory to DIR.  The default DIR is the value of the
HOME shell variable.  If DIR is "-", it is converted to $OLDPWD.

The variable CDPATH defines the search path for the directory containing
DIR.  Alternative directory names in CDPATH are separated by a colon (:).
A null directory name is the same as the current directory.  If DIR begins
with a slash (/), then CDPATH is not used.

If the directory is not found, and the shell option `cdable_vars' is set,
the word is assumed to be a variable name.  If that variable has a value,
its value is used for DIR.

Options:
  -L      force symbolic links to be followed: resolve symbolic
          links in DIR after processing instances of `..'
  -P      use the physical directory structure without following
          symbolic links: resolve symbolic links in DIR before
          processing instances of `..'
  -e      if the -P option is supplied, and the current working
          directory cannot be determined successfully, exit with
          a non-zero status
  -@      on systems that support it, present a file with extended
          attributes as a directory containing the file attributes

The default is to follow symbolic links, as if `-L' were specified.
`..' is processed by removing the immediately previous pathname component
back to a slash or the beginning of DIR.
```

Рисунок 10: Справка по команде cd


```
PWD(1)                                User Commands                                PWD(1)

NAME
    pwd - print name of current/working directory

SYNOPSIS
    pwd [OPTION]...

DESCRIPTION
    Print the full filename of the current working directory.

    -L, --logical
        use PWD from environment, even if it contains symlinks

    -P, --physical
        resolve all symlinks

    --help display this help and exit

    --version
        output version information and exit

    If no option is specified, -P is assumed.

    Your shell may have its own version of pwd, which usually supersedes the version described here.
    Please refer to your shell's documentation for details about the options it supports.

AUTHOR
    Written by Jim Meyering.

REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Report any translation bugs to <https://translationproject.org/team/>

Manual page pwd(1) line 1 (press h for help or q to quit)
```

Рисунок 11: Справка по команде pwd

```
MKDIR(1)                                User Commands                                MKDIR(1)

NAME
    mkdir - make directories

SYNOPSIS
    mkdir [OPTION]... DIRECTORY...

DESCRIPTION
    Create the DIRECTORY(ies), if they do not already exist.

    Mandatory arguments to long options are mandatory for short options too.

    -m, --mode=MODE
        set file mode (as in chmod), not a=rwx - umask

    -p, --parents
        no error if existing, make parent directories as needed, with their file modes unaffected
        by any -m option

    -v, --verbose
        print a message for each created directory

    -Z
        set SELinux security context of each created directory to the default type

    --context[=CTX]
        like -Z, or if CTX is specified then set the SELinux or SMACK security context to CTX

    --help display this help and exit

    --version
        output version information and exit

AUTHOR
    Manual page mkdir(1) line 1 (press h for help or q to quit)
```

Рисунок 12: Справка по команде mkdir

```
RMDIR(1)                                User Commands                                RMDIR(1)

NAME
    rmdir - remove empty directories

SYNOPSIS
    rmdir [OPTION]... DIRECTORY...

DESCRIPTION
    Remove the DIRECTORY(ies), if they are empty.

    --ignore-fail-on-non-empty
        ignore each failure to remove a non-empty directory

    -p, --parents
        remove DIRECTORY and its ancestors; e.g., 'rmdir -p a/b' is similar to 'rmdir a/b a'

    -v, --verbose
        output a diagnostic for every directory processed

    --help display this help and exit

    --version
        output version information and exit

AUTHOR
    Written by David MacKenzie.

REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Report any translation bugs to <https://translationproject.org/team/>

COPYRIGHT
    Copyright © 2025 Free Software Foundation, Inc. License GPLv3+: GNU GPL version 3 or later
    Manual page rmdir(1) line 1 (press h for help or q to quit)
```

Рисунок 13: Справка по команде rmdir

```
RM(1)                                User Commands                                RM(1)

NAME
  rm - remove files or directories

SYNOPSIS
  rm [OPTION]... [FILE]...

DESCRIPTION
  This manual page documents the GNU version of rm.  rm removes each specified file.  By default,
  it does not remove directories.

  If the -I or --interactive=once option is given, and there are more than three files or the -r,
-R, or --recursive are given, then rm prompts the user for whether to proceed with the entire op-
eration.  If the response is not affirmative, the entire command is aborted.

  Otherwise, if a file is unwritable, standard input is a terminal, and the -f or --force option is
not given, or the -i or --interactive=always option is given, rm prompts the user for whether to
remove the file.  If the response is not affirmative, the file is skipped.

OPTIONS
  Remove (unlink) the FILE(s).

  -f, --force
    ignore nonexistent files and arguments, never prompt

  -i
    prompt before every removal

  -I
    prompt once before removing more than three files, or when removing recursively; less in-
trusive than -i, while still giving protection against most mistakes

  --interactive[=WHEN]
    prompt according to WHEN: never, once (-I), or always (-i); without WHEN, prompt always

Manual page rm(1) line 1 (press h for help or q to quit)
```

Рисунок 14: Справка по команде rm

```
164  ls -a
165  ls -f
166  cd /var/spool/
167  ls -l
168  cd
169  ls
170  ls -al
171  mkdir newdir
172  mkdir newdir/morefun
173  mkdir letters memos misk
174  ls
175  rm letters/ memos/ misk/
176  rm -r letters/ memos/ misk/
177  rm -r newdir/
178  ls
179  ls -R
180  ls -t
181  help cd
182  man pwd
183  man mkdir
184  man rmdir
185  man rm
186  history
aaharlamova@aaharlamova:~$
```

Рисунок 15: Команда history

3. Выводы по проделанной работе

Мы приобрели практические навыки взаимодействия пользователя с системой посредством командной строки.